

Establishment of a Real-World Intelligent 5G Network Slicing and Mobility Research Testbed

Detailed Technical Assessment & Procurement Proposal

June 29, 2026

Contents

1	Executive Summary	5
2	Problem Statement & Motivations	5
3	3GPP 5G Standalone Core Network Architecture	6
3.1	5G Core Network Functions	6
3.2	Control Plane and User Plane Separation	6
3.3	Service-Based Architecture	7
3.4	Role of the 5G Core in Mobility Management	8
4	5G Network Interfaces and End-to-End Data Flow	9
4.1	Standardized 5G Interfaces	9
4.2	End-to-End Packet Flow	9
4.3	Control Plane versus User Plane Traffic	9
4.4	Importance of Standardized Interfaces in Mobility Research	10
5	5G Registration, Session Establishment and Mobility Call Flow	11
5.1	Phase I: Initial Registration	11
5.2	Phase II: PDU Session Establishment	12
5.3	Phase III: Continuous Mobility Monitoring	12
5.4	Phase IV: Intelligent Handover Procedure	13
5.5	Preserving IP Continuity During Mobility	13
6	AI-Assisted Mobility Management Framework	14
6.1	Overall AI Decision Pipeline	15
6.2	Feature Collection	15
6.3	Prediction Module	15
6.4	Reinforcement Learning Based Decision Engine	16
6.5	Closed Loop Network Optimization	17

7	Experimental Methodology and Evaluation Framework	17
7.1	Experimental Workflow	18
7.2	Baseline Mobility Algorithms	18
7.3	Mobility Scenarios	18
7.4	Traffic Profiles	18
7.5	Performance Metrics	19
7.6	Data Collection Pipeline	19
7.7	Statistical Analysis	20
7.8	Expected Deliverables	21
8	Laboratory Infrastructure and Physical Deployment Design	21
8.1	Overall Laboratory Architecture	22
8.2	Hardware Rack Organization	22
8.3	Radio Access Network Deployment	22
8.4	UE Mobility Corridor	23
8.5	Network Backbone	24
8.6	Power and Cooling Infrastructure	24
8.7	Future Expansion Capability	24
9	Research Objectives and Planned Experimental Campaign	25
9.1	Research Objective 1: Intelligent Mobility Prediction	25
9.2	Research Objective 2: AI-Assisted Handover Optimization	26
9.3	Research Objective 3: Intelligent Network Slice Selection	26
9.4	Research Objective 4: Edge Service Continuity	27
9.5	Research Objective 5: Energy-Efficient Mobility Management	27
9.6	Experimental Campaign	27
9.7	Expected Scientific Contributions	28
9.8	Potential Research Outcomes	28
10	Detailed Hardware Ecosystem of the Proposed 5G Testbed	28
10.1	Hardware Classification	29
10.2	5G Core Infrastructure	29
10.3	Radio Access Network	29
10.4	Commercial User Equipments	30
10.5	Edge Computing Infrastructure	30
10.6	AI Computing Infrastructure	31
10.7	Monitoring Infrastructure	31
10.8	Network Backbone	32
10.9	Synchronization Infrastructure	32
10.10	Power Infrastructure	32
10.11	Hardware Interaction	33
10.12	Scalability Considerations	33

11 Proposed Testbed Architecture	34
11.1 Logical Architecture	34
11.2 End-to-End UE Registration Flow	34
11.3 Physical Deployment Topology	34
12 System Design and End-to-End Operational Architecture	35
12.1 Layered System Architecture	35
12.2 Operational Workflow	35
12.3 Closed-Loop System Operation	36
12.4 End-to-End Data Flow	36
12.5 Component Interaction Matrix	36
12.6 Data Lifecycle	38
12.7 System Design Advantages	38
13 Technical Justification of the Proposed Infrastructure	39
13.1 5G Core Server	39
13.2 Software Defined Radios	39
13.3 Commercial User Equipments	39
13.4 Multi-access Edge Computing Server	40
13.5 Artificial Intelligence Workstation	40
13.6 Monitoring Server	40
13.7 Network Attached Storage	40
13.8 Networking Infrastructure	40
13.9 Power Infrastructure	41
13.10Infrastructure Integration	41
13.11Overall Engineering Rationale	41
14 Deployment Strategy and System Integration	42
14.1 Phase I: Laboratory Preparation	42
14.2 Phase II: Compute Infrastructure Deployment	42
14.3 Phase III: Network Infrastructure Deployment	43
14.4 Phase IV: 5G Core Deployment	43
14.5 Phase V: Radio Access Network Deployment	44
14.6 Phase VI: Monitoring and Telemetry Deployment	44
14.7 Phase VII: AI Framework Deployment	44
14.8 System Integration Workflow	45
14.9 Verification Checklist	45
14.10Expected Deployment Duration	46
15 Required Modules and Procurement Matrix	46

16 Detailed Module Technical Specifications & Visualizations	48
16.1 Module 1: Software-Defined Radio (SDR) RAN Transceiver	48
16.2 Module 2: 5G Core Network Rack Server	49
16.3 Module 3: AI and Mobility Controller Workstation	50
16.4 Module 4: Central Storage NAS Chassis	50
17 Software Stack Integration Plan	52
18 Junior Research Fellow (JRF) Profile	52
18.1 Essential Technical Qualifications	53
18.2 Desirable Technical Qualifications	53
19 Risk Assessment & Mitigation Plan	53

1 Executive Summary

This proposal details the requirements, architectural specifications, procurement sources, and budget estimates for establishing a physical, real-world 5G Network Slicing and Mobility Research Testbed. The testbed will serve as a high-fidelity platform for developing, validating, and evaluating Deep Neural Network (DNN) and Reinforcement Learning (RL) based slice-switching, mobility management, and intelligent handover algorithms.

Traditional simulation software fails to capture real-world RF channel variations, hardware latencies, network stack overheads, and dynamic hardware limits. Consequently, a physical deployment is required. This testbed consists of a 5G Standalone (SA) Core Network, Software-Defined Radio (SDR) based Radio Access Network (RAN), commercial 5G User Equipments (UEs), Multi-access Edge Computing (MEC) servers, a dedicated AI inference workstation, network telemetry infrastructure, and appropriate power backups.

The technical specifications outlined in this document are aligned with standard vendor availability in India. In addition to hardware specifications and costs, this report lists verified procurement sources (e.g., GSAS Micro Systems, ServerBasket, Computech, Volted PC) and includes architectural diagrams and product visualizations to facilitate rapid administrative approval and procurement.

2 Problem Statement & Motivations

5G Network Slicing allows operators to partition a single physical network into multiple virtual networks tailored to specific Quality of Service (QoS) requirements (e.g., Ultra-Reliable Low-Latency Communication [URLLC], Enhanced Mobile Broadband [eMBB], and Massive Machine-Type Communication [mMTC]). However, maintaining service continuity and meeting SLA (Service Level Agreement) requirements during user mobility presents significant research challenges:

1. **Slice-Blind Handover Mechanisms:** Standard 3GPP handover procedures focus primarily on radio signal strength (RSRP/RSRQ) and do not account for application-level slice requirements or edge compute server load. As a result, a UE may be handed over to a gNodeB that cannot support its active slice SLAs.**Dynamic Resource Renegotiation:** Mobile users traversing cells require seamless, real-time slice reallocation and slice boundary adjustments. Static slicing leads to either resource starvation or highly inefficient over-provisioning.
2. **Edge Service Redirection Latency:** In a Multi-access Edge Computing (MEC) deployment, when a UE performs a handover to a new gNodeB, its corresponding application container at the edge must migrate or redirect traffic. If the redirection is not coordinated with the radio-layer handover, packet loss and latency spikes occur.

To solve these problems, researchers require a physical 5G Standalone environment that allows modification of the Core Network (using Open5GS) and the RAN stack (using srsRAN or OpenAirInterface). This testbed will allow developers to deploy and test intelligent, AI-assisted controllers that intercept control-plane messages, predict user trajectories, pre-allocate slice resources, and trigger pro-active handovers.

3 3GPP 5G Standalone Core Network Architecture

The proposed research testbed follows the 3GPP Release-16 Service Based Architecture (SBA) for a 5G Standalone (SA) deployment. Unlike previous LTE EPC architectures that relied upon tightly coupled network entities, the 5G Core decomposes network functionality into modular Network Functions (NFs) communicating through standardized Service-Based Interfaces (SBIs). This modular architecture allows researchers to independently modify mobility management, policy control, authentication, session establishment, and user-plane forwarding without affecting the remaining components of the network.

The adoption of a standards-compliant 5G Core enables the proposed testbed to emulate realistic operator deployments while providing sufficient flexibility for implementing AI-assisted mobility prediction, dynamic network slicing, edge service migration, and intelligent handover optimization algorithms.

3.1 5G Core Network Functions

The principal network functions implemented within the proposed testbed are summarized in Table ???. Together, these entities provide complete subscriber authentication, mobility management, session establishment, policy enforcement, and user-plane routing required for a fully operational private 5G Standalone network.

3.2 Control Plane and User Plane Separation

One of the defining characteristics of the 5G Standalone architecture is the complete separation between the Control Plane (CP) and the User Plane (UP). Such separation allows independent scaling of signalling functions and high-throughput packet forwarding, thereby improving network flexibility and reducing latency.

The Control Plane manages signalling messages, subscriber authentication, mobility procedures, policy enforcement, and session establishment, whereas the User Plane exclusively processes user traffic. This architectural decoupling enables AI-based mobility controllers to intercept signalling events without introducing unnecessary overhead into user data forwarding.

illustrates the logical separation adopted within the proposed testbed.

Table 1: Major 5G Core Network Functions

Network Function	Primary Responsibilities
AMF	Access and Mobility Management Function responsible for UE registration, authentication coordination, mobility management, NAS signalling, security context establishment, and handover orchestration.
SMF	Session Management Function responsible for PDU Session establishment, IP address allocation, QoS management, UPF selection, tunnel management, and interaction with the User Plane Function.
UPF	User Plane Function responsible for packet forwarding, GTP-U encapsulation/decapsulation, traffic steering, QoS enforcement, buffering, and routing packets toward external Data Networks.
UDM	Unified Data Management stores subscriber information including IMSI, authentication credentials, subscribed slices, QoS profiles, and subscription data.
AUSF	Authentication Server Function performs primary authentication procedures using subscriber credentials provided by the UDM.
NRF	Network Repository Function maintains service registration of all Network Functions and provides service discovery within the Service-Based Architecture.
PCF	Policy Control Function generates policy decisions regarding QoS, charging, traffic prioritization, and slice-specific resource allocation.
NSSF	Network Slice Selection Function determines the appropriate Network Slice Instance (NSI) according to subscriber subscription and service requirements.

3.3 Service-Based Architecture

Unlike LTE EPC, the 5G Core follows a Service-Based Architecture (SBA) in which every Network Function exposes RESTful services over HTTP/2. Instead of establishing static peer-to-peer relationships, each Network Function dynamically discovers other services through the Network Repository Function (NRF). This architecture significantly improves modularity, scalability, and service interoperability.

Within the proposed research platform, the Service-Based Architecture provides an ideal environment for integrating Artificial Intelligence modules into the signalling path. The AI Mobility Controller can subscribe to mobility events generated by the AMF, retrieve network statistics from the SMF and UPF, and dynamically influence mobility decisions without requiring modifications to the underlying transport infrastructure.

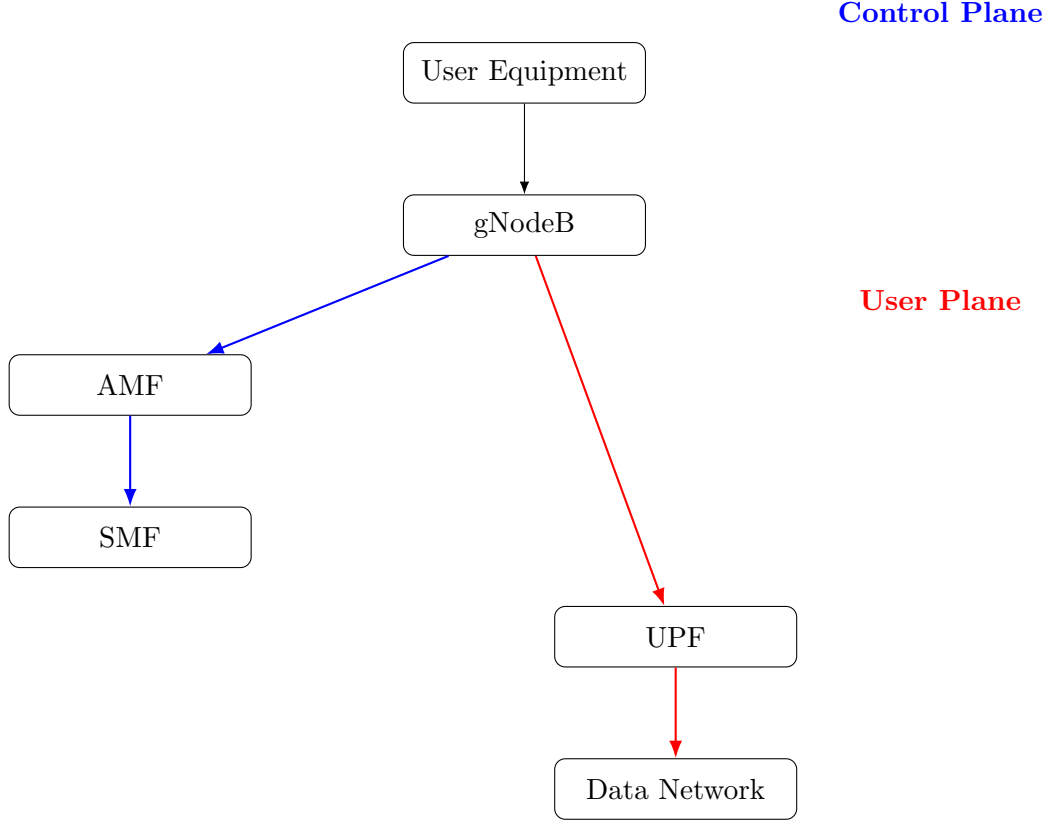


Figure 1: Logical Separation of the Control Plane and User Plane in a 5G Standalone Network

3.4 Role of the 5G Core in Mobility Management

The primary objective of this research project is the investigation of intelligent mobility management under highly dynamic network conditions. During a handover procedure, multiple network functions cooperate to ensure uninterrupted connectivity.

The AMF maintains the UE context and coordinates handover signalling between neighbouring gNodeBs. The SMF updates session information and modifies GTP-U tunnels whenever the serving cell changes. Simultaneously, the UPF redirects user traffic toward the newly selected radio access node while preserving the user's allocated IP address. Subscriber authentication and slice eligibility continue to be verified using the UDM, AUSF, and NSSF throughout the mobility procedure.

This coordinated interaction among network functions enables seamless user mobility while maintaining Quality of Service (QoS), minimizing packet loss, and preserving application sessions. The proposed AI-assisted mobility controller extends this conventional architecture by predicting future handover events, proactively reserving network resources, and optimizing slice allocation before the physical handover occurs.

4 5G Network Interfaces and End-to-End Data Flow

The interaction among various Network Functions (NFs) in a 5G Standalone (SA) deployment is governed by standardized interfaces defined by the 3GPP specifications. These interfaces enable secure communication between the User Equipment (UE), the Radio Access Network (RAN), the 5G Core (5GC), and external Data Networks (DN). Understanding these interfaces is essential for designing mobility-aware algorithms, since every handover event, session establishment, and network slice transition propagates through one or more of these communication channels.

Unlike LTE, where signalling was tightly coupled to fixed interfaces, the 5G Core adopts a Service-Based Architecture (SBA), allowing Network Functions to dynamically discover and communicate with one another while maintaining standardized external interfaces toward the RAN and User Equipment.

4.1 Standardized 5G Interfaces

Table ?? summarizes the principal interfaces implemented within the proposed research testbed.

4.2 End-to-End Packet Flow

The complete communication path between a User Equipment (UE) and an external application server traverses multiple logical interfaces within the 5G architecture.

Initially, the UE establishes radio connectivity with the serving gNodeB. Registration signalling is transmitted through the N2 interface toward the Access and Mobility Management Function (AMF), where authentication and mobility procedures are initiated. Upon successful authentication, the Session Management Function (SMF) allocates a PDU session and configures the User Plane Function (UPF). User packets are subsequently encapsulated using the GTP-U protocol and forwarded through the N3 interface toward the UPF. Finally, traffic exits the 5G Core via the N6 interface toward the target Data Network or Internet.

The proposed research testbed inserts AI-assisted decision modules within this signalling path to monitor network conditions and optimize mobility decisions without disrupting user-plane forwarding.

4.3 Control Plane versus User Plane Traffic

The 5G architecture distinguishes signalling traffic from user data traffic to improve scalability and simplify network management.

- **Control Plane (Blue Path):** Carries signalling information required for authentication, registration, mobility management, network slice selection, QoS negotiation,

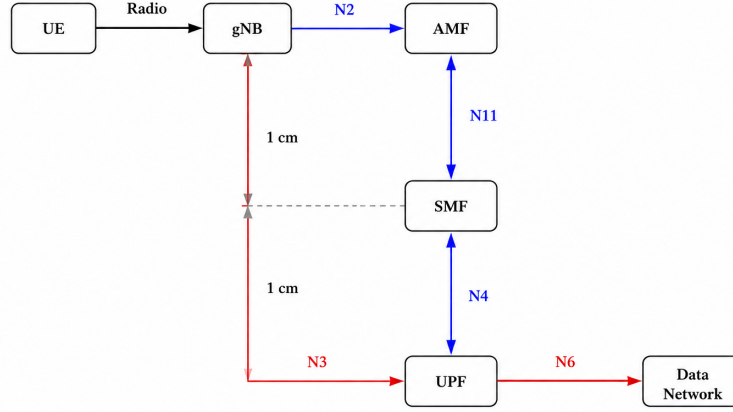


Fig. 1: End-to-End Control Plane and User Plane Communication Flow.

Figure 2:

and session establishment. This traffic primarily flows between the UE, gNodeB, AMF, SMF, and associated Core Network Functions.

- **User Plane (Red Path):** Carries application data generated by users, including web browsing, video streaming, voice traffic, IoT sensor data, and edge computing traffic. User-plane packets are forwarded directly between the gNodeB and the UPF before reaching external Data Networks.

The proposed AI-assisted mobility controller primarily interacts with the Control Plane by monitoring signalling events generated by the AMF and SMF while simultaneously observing User Plane Quality of Service (QoS) metrics collected from the UPF. This separation allows mobility decisions to be optimized without introducing additional latency into user traffic.

4.4 Importance of Standardized Interfaces in Mobility Research

Each interface plays a critical role during handover procedures. As the User Equipment moves between neighbouring cells, measurement reports generated by the UE are transported through the N2 interface toward the AMF. The AMF coordinates with neighbouring gNodeBs via the Xn interface (or indirectly through the Core Network when Xn is unavailable), while the SMF updates forwarding policies within the UPF through the N4 interface.

Because the user's IP address remains anchored at the UPF, intelligent path switching can be achieved without interrupting active application sessions. This characteristic forms the foundation for implementing predictive mobility management algorithms, AI-assisted slice migration, proactive resource reservation, and seamless service continuity within the proposed research platform.

5 5G Registration, Session Establishment and Mobility Call Flow

The successful operation of a 5G Standalone (SA) network depends upon a sequence of standardized control-plane procedures that enable a User Equipment (UE) to authenticate itself, establish an IP session, and maintain uninterrupted connectivity while moving across multiple radio cells. These procedures collectively define the lifecycle of a UE from initial power-on to continuous mobility.

The proposed research testbed closely follows the 3GPP Release-16 call flow while introducing Artificial Intelligence (AI) assisted decision modules capable of predicting user mobility, optimizing slice allocation, and minimizing handover latency.

5.1 Phase I: Initial Registration

When a User Equipment (UE) powers on, it first searches for available Synchronization Signal Blocks (SSBs) transmitted by nearby gNodeBs. Based on the received signal quality (RSRP, RSRQ and SINR), the UE selects the most suitable serving cell and establishes an initial Radio Resource Control (RRC) connection.

Following radio synchronization, the UE initiates the NAS Registration procedure through the gNodeB. The Access and Mobility Management Function (AMF) validates the subscriber identity with the Authentication Server Function (AUSF) and Unified Data Management (UDM), establishes a security context, and registers the UE within the Core Network.

illustrates the complete registration procedure.

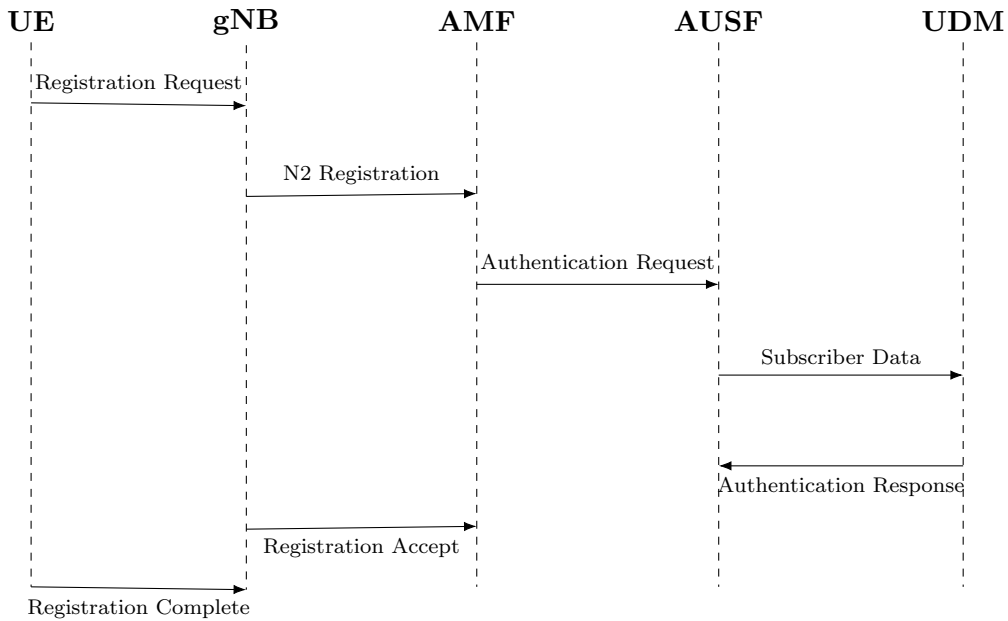


Figure 3: Simplified 5G UE Registration Message Sequence.

5.2 Phase II: PDU Session Establishment

Following successful authentication, the UE requests establishment of a Packet Data Unit (PDU) session. The AMF forwards the request to the Session Management Function (SMF), which allocates an IP address, selects an appropriate User Plane Function (UPF), and configures forwarding rules through the PFCP protocol.

After successful tunnel establishment, GTP-U tunnels are created between the serving gNodeB and the UPF. User traffic can now reach external Data Networks through the N6 interface.

The resulting communication path is illustrated in .

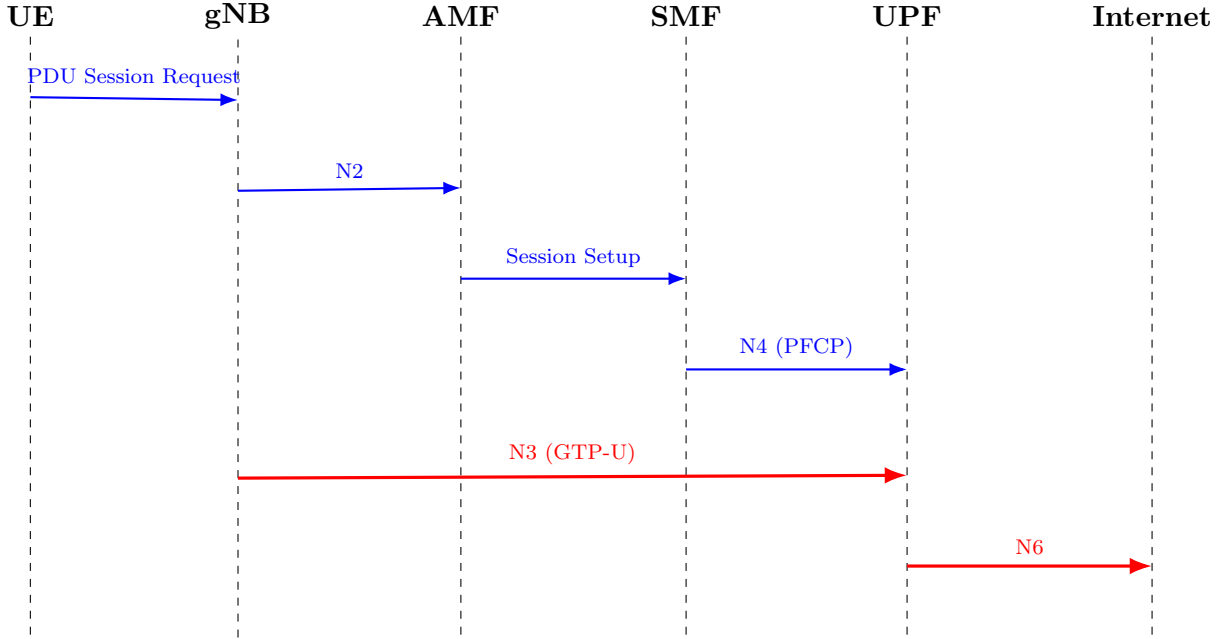


Figure 4: Simplified PDU Session Establishment Procedure Showing Control and User Plane Interfaces.

5.3 Phase III: Continuous Mobility Monitoring

Once connected, the UE continuously measures neighbouring cells using Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), Signal-to-Interference-plus-Noise Ratio (SINR), and Channel Quality Indicator (CQI).

These measurements are periodically transmitted to the serving gNodeB through Measurement Reports.

The gNodeB forwards the collected measurements to the AMF, enabling the Core Network to evaluate whether a handover should be initiated.

Within the proposed research platform, these measurement reports are simultaneously streamed to the AI Mobility Controller, where Deep Neural Networks and Reinforcement Learning models estimate future user trajectories and proactively prepare neighbouring cells before signal degradation occurs.

5.4 Phase IV: Intelligent Handover Procedure

Conventional 5G handover decisions are largely based on signal quality thresholds and hysteresis timers. Although effective for maintaining radio connectivity, these methods are unaware of network slice availability, edge computing load, application latency requirements, and future user movement.

The proposed research introduces an AI-assisted mobility framework capable of evaluating multiple network parameters simultaneously before initiating a handover.

The complete handover process is summarized below:

1. UE periodically transmits Measurement Reports.
2. AI Mobility Controller predicts future trajectory.
3. Candidate gNodeBs are ranked according to:
 - Signal strength
 - Slice availability
 - MEC utilization
 - Current traffic load
 - Predicted congestion
 - User QoS requirements
4. Target gNodeB resources are reserved.
5. AMF initiates handover.
6. SMF updates forwarding policies.
7. UPF redirects GTP-U tunnels.
8. UE synchronizes with the target gNodeB.
9. Source gNodeB releases radio resources.

5.5 Preserving IP Continuity During Mobility

One of the principal objectives of this research is to preserve the UE's allocated IP address throughout inter-cell mobility. Since the User Plane Function (UPF) acts as the IP anchor within the 5G Core, changes in the serving gNodeB do not require IP reassignment.

Instead, only the GTP-U forwarding tunnels are modified while maintaining the same logical IP session. Consequently, ongoing TCP sessions, multimedia streams, edge computing applications, and network slice contexts remain uninterrupted during mobility.

The proposed AI framework further reduces handover interruption time by proactively configuring forwarding paths before the UE reaches the target cell, thereby minimizing packet loss, latency spikes, and service disruption.

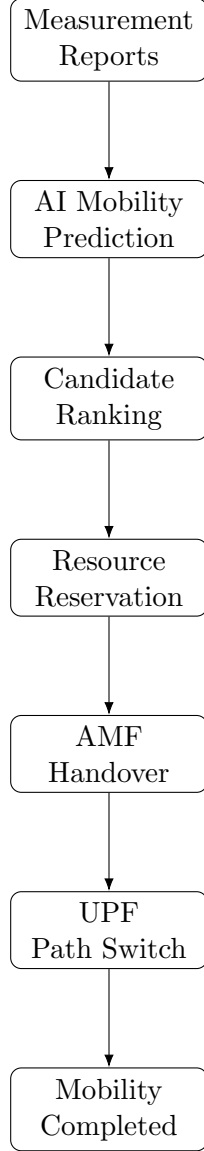


Figure 5: AI-Assisted Intelligent Handover Procedure

6 AI-Assisted Mobility Management Framework

Conventional mobility management algorithms defined by the 3GPP specifications primarily rely upon radio measurements such as the Reference Signal Received Power (RSRP), Reference Signal Received Quality (RSRQ), Signal-to-Interference-plus-Noise Ratio (SINR), and configurable hysteresis timers. While these approaches are effective for maintaining basic radio connectivity, they remain unaware of higher-layer network conditions including edge server utilization, slice availability, Quality of Service (QoS) requirements, traffic congestion, and predicted user trajectories.

Consequently, users may be handed over to geographically closer cells that are already overloaded, lack the required network slice resources, or host heavily loaded edge computing servers. Such decisions often lead to unnecessary packet loss, increased latency, degraded Quality of Experience (QoE), and frequent handover failures.

The proposed research introduces an Artificial Intelligence assisted mobility framework capable of integrating radio measurements with network-wide telemetry to perform predictive and context-aware mobility management.

6.1 Overall AI Decision Pipeline

The proposed framework continuously collects measurements from multiple layers of the network.

- User Equipment (UE) radio measurements
- gNodeB scheduling statistics
- Core Network signaling events
- User Plane traffic statistics
- Edge Computing resource utilization
- Network Slice resource allocation
- Historical mobility traces

These heterogeneous data streams are synchronized and processed by a centralized AI Mobility Controller responsible for predicting future network conditions and selecting the optimal target cell before conventional threshold-based handover algorithms are triggered.

6.2 Feature Collection

Unlike conventional mobility algorithms that only utilize signal strength, the proposed framework constructs a high-dimensional feature vector by aggregating measurements from multiple network entities.

The collected parameters include

6.3 Prediction Module

The prediction engine estimates the future state of the network several seconds ahead.

Instead of reacting after signal degradation has already occurred, the AI controller predicts

- Future UE trajectory
- Future serving cell
- Expected signal quality
- Candidate gNodeB load

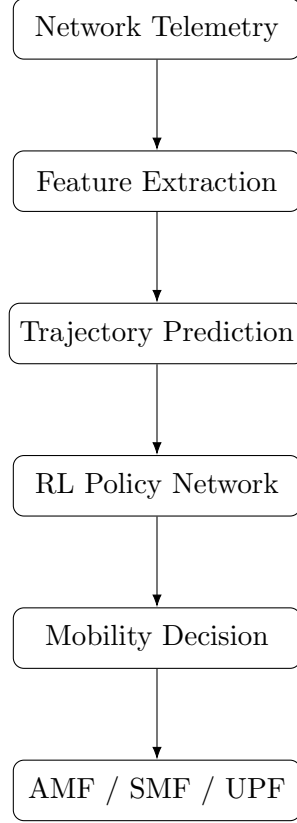


Figure 6: Overall AI-Assisted Mobility Decision Pipeline

- Slice availability
- MEC server congestion
- Network latency
- Probability of handover failure

These predictions enable proactive preparation of neighboring cells before the UE physically reaches the cell boundary.

6.4 Reinforcement Learning Based Decision Engine

A Deep Reinforcement Learning (DRL) agent continuously interacts with the network environment to determine the optimal mobility action.

The state space consists of the complete network feature vector collected from the Radio Access Network, Core Network, and Edge Computing infrastructure.

The action space includes

- Maintain current serving cell
- Trigger handover
- Delay handover

- Reserve neighboring resources
- Switch Network Slice
- Migrate MEC application
- Modify QoS policy

The reward function is designed to maximize

- Successful handovers
- End-to-end throughput
- Slice utilization
- User Quality of Experience (QoE)

while minimizing

- Packet loss
- Latency
- Ping-pong handovers
- Energy consumption
- Signaling overhead

6.5 Closed Loop Network Optimization

Unlike traditional reactive mobility management, the proposed framework forms a continuous feedback loop.

7 Experimental Methodology and Evaluation Framework

The primary objective of the proposed testbed is to experimentally evaluate Artificial Intelligence assisted mobility management algorithms under realistic 5G Standalone deployment scenarios. Unlike simulation-only studies, the proposed infrastructure enables controlled experimentation over real radio hardware, commercial User Equipments (UEs), physical servers, and software-defined radio base stations.

The evaluation methodology has been designed to ensure reproducibility, statistical significance, and objective comparison against conventional 3GPP mobility management algorithms.

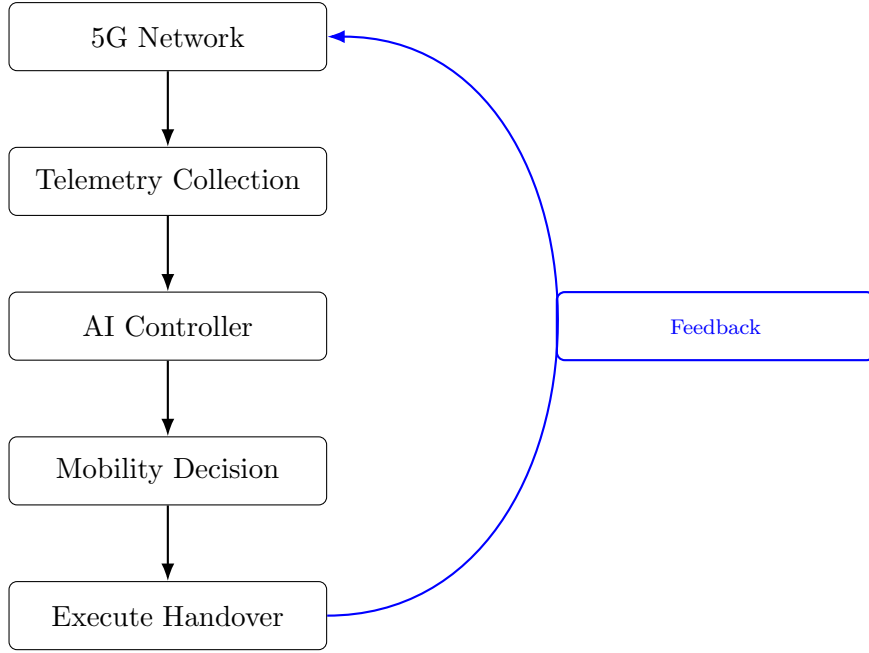


Figure 7: Closed-Loop AI Mobility Optimization Framework.

7.1 Experimental Workflow

Each experiment follows a standardized workflow beginning with network initialization and ending with statistical performance analysis.

7.2 Baseline Mobility Algorithms

The proposed AI framework will be compared against conventional mobility management techniques implemented within current 3GPP-compliant cellular systems.

The following baseline algorithms will be evaluated.

7.3 Mobility Scenarios

To evaluate algorithm robustness under varying operating conditions, multiple mobility scenarios will be investigated.

7.4 Traffic Profiles

The network will be subjected to multiple application workloads representative of practical deployments.

- UHD Video Streaming
- VoNR Voice Calls
- Web Browsing

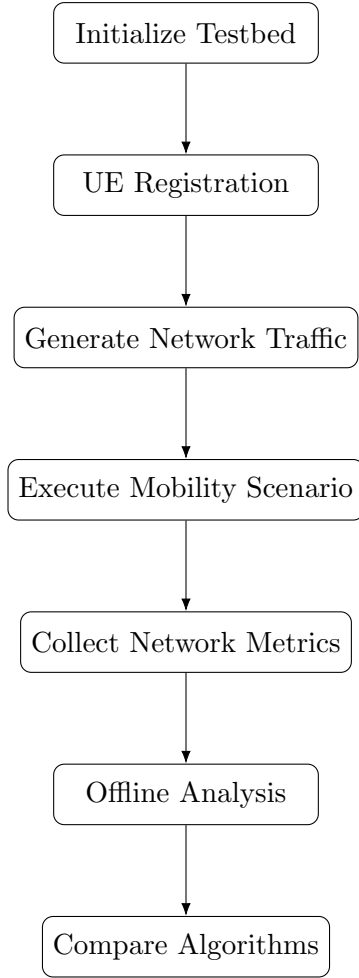


Figure 8: Overall Experimental Workflow

- IoT Telemetry
- MEC Video Analytics
- Large File Transfer
- Mixed Enterprise Traffic

7.5 Performance Metrics

Each experiment will record both radio-level and network-level performance indicators.

7.6 Data Collection Pipeline

Experimental measurements will be collected simultaneously from multiple infrastructure components.

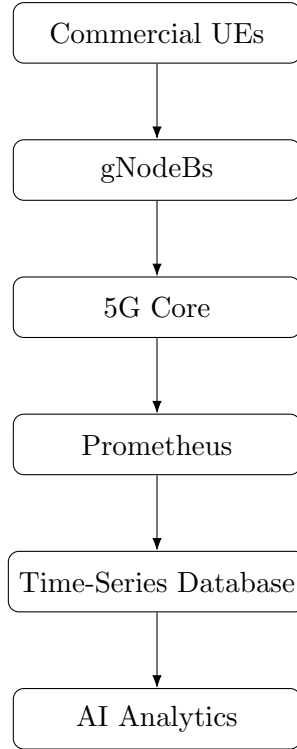


Figure 9: Experimental Data Collection Pipeline

7.7 Statistical Analysis

Each experiment will be repeated multiple times to eliminate random fluctuations caused by radio propagation, operating system scheduling, and wireless interference.

The collected measurements will be statistically analyzed using

- Mean
- Median
- Standard Deviation
- Confidence Interval
- Box Plots
- Cumulative Distribution Function (CDF)
- One-way ANOVA
- Paired t-test

Only statistically significant improvements will be considered valid research outcomes.

7.8 Expected Deliverables

The proposed experimental campaign is expected to generate

- Large-scale mobility datasets
- AI-ready training datasets
- Open-source mobility framework
- Performance benchmark reports
- Conference publications
- Journal publications
- Technology demonstrators
- Potential patent filings

The resulting datasets and performance evaluations will provide quantitative evidence of the effectiveness of AI-assisted mobility management under realistic deployment conditions.

8 Laboratory Infrastructure and Physical Deployment Design

The proposed research facility is designed as a compact private 5G Standalone laboratory capable of supporting realistic mobility experiments involving multiple radio cells, commercial User Equipments (UEs), edge computing servers, AI-assisted mobility controllers, and a standards-compliant 5G Core Network.

Unlike purely software-based simulators, the proposed deployment incorporates physical Software Defined Radios (SDRs), dedicated compute servers, commercial networking equipment, and programmable User Equipments to accurately reproduce real-world radio propagation, network latency, and mobility behaviour.

The laboratory is organized into five major infrastructure layers:

1. Core Network Infrastructure
2. Radio Access Network (RAN)
3. Edge Computing Infrastructure
4. AI and Monitoring Infrastructure
5. Network Backbone and Power Distribution

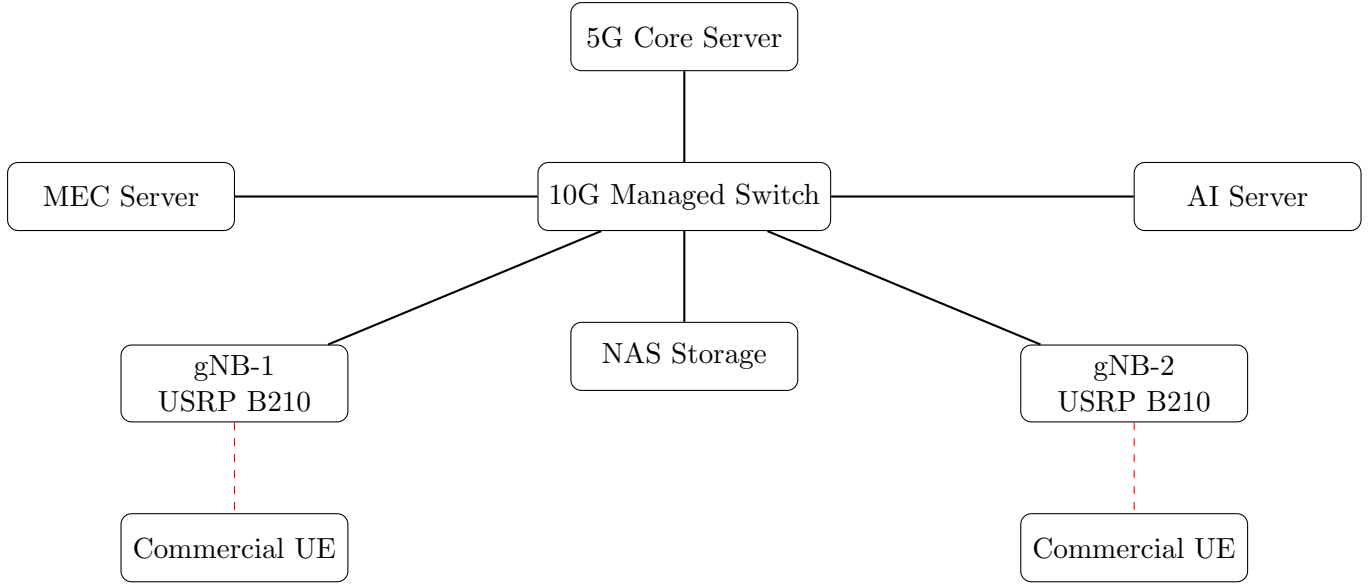


Figure 10: Overall Laboratory Infrastructure

8.1 Overall Laboratory Architecture

illustrates the complete laboratory deployment.

8.2 Hardware Rack Organization

To improve maintainability, thermal management, and cable routing, all compute equipment will be installed within a standard 42U server rack.

The proposed rack layout is shown in .

8.3 Radio Access Network Deployment

The Radio Access Network consists of two Software Defined Radio based gNodeBs positioned to create overlapping coverage regions. This arrangement enables continuous mobility experiments involving inter-cell handovers.

Each gNodeB comprises

- Ettus USRP B210 SDR
- Dedicated Host Computer
- External MIMO Antennas
- GPSDO (optional future upgrade)
- RF Attenuators for indoor testing

The overlapping coverage enables controlled execution of

- Intra-frequency handover

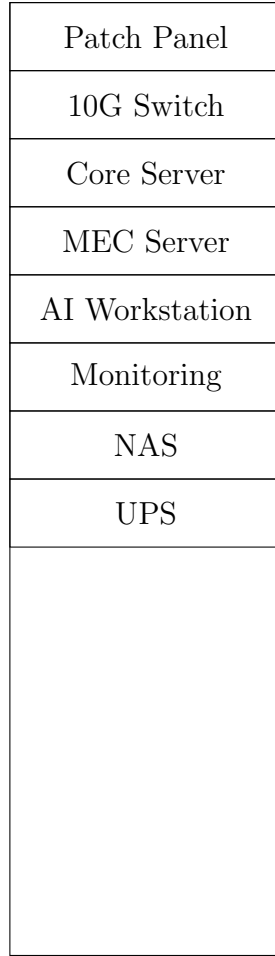


Figure 11: Proposed Server Rack Organization

- Inter-cell mobility
- Slice migration
- Edge service migration
- Packet loss analysis

8.4 UE Mobility Corridor

To reproduce realistic mobility, a dedicated experimental corridor will be established inside the laboratory.



Figure 12: Indoor Mobility Corridor

The mobility corridor enables repeatable experiments at controlled pedestrian speeds while ensuring identical trajectories across multiple experimental trials.

8.5 Network Backbone

All infrastructure components communicate through a centralized Layer-2/Layer-3 managed switch supporting both Gigabit Ethernet and 10 Gigabit SFP+ uplinks.

The proposed backbone supports

- 10 Gbps Core Server connectivity
- MEC communication
- NAS storage
- Monitoring traffic
- AI telemetry
- SDR host computers

This architecture minimizes bottlenecks during high-throughput mobility experiments.

8.6 Power and Cooling Infrastructure

Continuous operation of the testbed requires reliable electrical and thermal management.

The laboratory includes

- 5 kVA Online UPS
- Rack-mounted Power Distribution Unit
- Surge Protection
- Active Rack Cooling
- Dedicated Air Conditioning
- Automatic Shutdown Scripts

These provisions ensure uninterrupted execution of long-duration experiments and protect critical infrastructure against power failures.

8.7 Future Expansion Capability

Although initially designed as a two-cell deployment, the proposed infrastructure has been intentionally over-engineered to accommodate future research activities.

The laboratory can be expanded to support

- Four or more SDR-based gNodeBs
- Open RAN deployments

- O-RAN Near-RT RIC
- Multi-UPF architectures
- Additional MEC clusters
- AI-native 6G experimentation
- Satellite Non-Terrestrial Networks (NTN)
- Large-scale Network Slice orchestration

Consequently, the proposed laboratory serves not only the objectives of the current research project but also establishes a long-term experimental platform for future wireless communication research.

9 Research Objectives and Planned Experimental Campaign

The establishment of the proposed 5G Standalone research testbed enables a comprehensive investigation into intelligent mobility management, AI-assisted network slicing, and edge-aware service orchestration under realistic deployment conditions. Rather than serving solely as an infrastructure project, the laboratory has been designed to facilitate long-term experimental research aligned with current challenges in beyond-5G (B5G) and emerging 6G communication systems.

The experimental campaign is organized into multiple research themes, each targeting a specific aspect of intelligent mobility management and network optimization.

9.1 Research Objective 1: Intelligent Mobility Prediction

The first objective focuses on predicting future user mobility before conventional handover triggers are activated.

Current cellular systems initiate handovers only after radio signal quality begins to degrade. Such reactive mechanisms often lead to increased packet loss, unnecessary signaling, and temporary degradation in Quality of Service (QoS).

The proposed research investigates machine learning models capable of predicting

- Future user trajectory
- Future serving cell
- Expected signal quality
- Probability of handover
- Time until handover

These predictions enable proactive resource reservation and intelligent mobility planning.

9.2 Research Objective 2: AI-Assisted Handover Optimization

The second research objective evaluates reinforcement learning algorithms capable of dynamically selecting the optimal target cell.

Unlike traditional strongest-signal approaches, the proposed framework simultaneously considers

- Radio signal quality
- Cell congestion
- Available bandwidth
- Edge server utilization
- Network slice availability
- Historical mobility information
- User Quality of Experience

The effectiveness of the proposed decision framework will be evaluated against standard 3GPP handover algorithms.

9.3 Research Objective 3: Intelligent Network Slice Selection

Network slicing represents one of the defining technologies of fifth-generation communication systems.

Instead of assigning users to static slices, this work investigates adaptive slice allocation strategies capable of responding to changing network conditions.

The proposed AI controller continuously evaluates

- Slice utilization
- SLA compliance
- User application requirements
- Available radio resources
- Edge computing availability

before selecting the most appropriate slice instance.

9.4 Research Objective 4: Edge Service Continuity

Modern low-latency applications frequently execute on Multi-access Edge Computing (MEC) servers.

During mobility, users may move away from the serving MEC node, increasing application latency.

This research investigates

- Dynamic edge service migration
- Container relocation
- Service redirection
- Traffic steering
- Edge resource prediction

to maintain uninterrupted application performance throughout mobility.

9.5 Research Objective 5: Energy-Efficient Mobility Management

In addition to maximizing Quality of Service, future communication networks must reduce infrastructure energy consumption.

The proposed framework investigates energy-aware mobility decisions considering

- Base station utilization
- Computational load
- Network traffic
- AI inference overhead
- Power consumption of SDR infrastructure

The resulting algorithms are expected to improve both network performance and operational efficiency.

9.6 Experimental Campaign

The experimental evaluation will be divided into multiple phases as summarized in Table ??.

9.7 Expected Scientific Contributions

The proposed research is expected to contribute

- AI-assisted mobility management algorithms.
- Reinforcement learning based handover optimization.
- Dynamic network slice selection framework.
- Edge-aware mobility orchestration.
- Large-scale real-world mobility datasets.
- Benchmark datasets for future wireless communication research.
- Open-source software components.
- Publications in IEEE Transactions, Elsevier, and Springer journals.
- Demonstration platform for future 6G research.

9.8 Potential Research Outcomes

Successful completion of the proposed work is expected to result in

- 3–5 IEEE conference publications.
- 2–3 SCI indexed journal publications.
- Open-source mobility framework.
- Patent applications.
- Master’s and PhD research support.
- National and international research collaborations.
- Future extensions toward AI-native 6G networks.

10 Detailed Hardware Ecosystem of the Proposed 5G Testbed

The proposed laboratory consists of multiple tightly integrated hardware subsystems that collectively emulate a realistic private 5G Standalone deployment. Unlike software-only simulations, the physical implementation introduces practical engineering considerations such as RF propagation, synchronization, packet forwarding latency, cable losses, thermal management, and hardware resource limitations.

The hardware ecosystem has been designed according to 3GPP Release-16 recommendations while remaining scalable for future Open RAN (O-RAN) and Beyond-5G (B5G) research activities.

10.1 Hardware Classification

The complete laboratory infrastructure is divided into ten major hardware domains.

10.2 5G Core Infrastructure

The Core Network forms the central intelligence of the private cellular network.

The dedicated Core Server hosts

- Access and Mobility Management Function (AMF)
- Session Management Function (SMF)
- User Plane Function (UPF)
- Unified Data Management (UDM)
- Authentication Server Function (AUSF)
- Network Repository Function (NRF)
- Policy Control Function (PCF)
- Network Slice Selection Function (NSSF)
- MongoDB Subscriber Database

This server is responsible for subscriber authentication, mobility management, session establishment, IP allocation, Quality of Service enforcement, and network slice orchestration.

10.3 Radio Access Network

The Radio Access Network is implemented using Ettus USRP B210 Software Defined Radios.

Each SDR is connected to a dedicated Linux workstation executing the software gNodeB stack.

Every gNodeB consists of

- Ettus USRP B210
- Intel-based Host Computer
- UHD Driver
- External MIMO Antennas
- RF Cabling

- SMA Connectors
- Variable RF Attenuators
- Power Supply

The two gNodeBs generate overlapping radio coverage necessary for executing mobility experiments.

10.4 Commercial User Equipments

Unlike software simulators, the proposed testbed employs commercial 5G smartphones supporting Standalone (SA) mode.

These devices execute

- Initial Registration
- Authentication
- PDU Session Establishment
- Mobility
- Slice Selection
- Application Traffic

The use of commercial devices enables realistic experimentation under practical radio conditions.

10.5 Edge Computing Infrastructure

A dedicated MEC server hosts latency-sensitive applications.

Typical workloads include

- Video Analytics
- Object Detection
- AI Inference
- IoT Gateways
- Containerized Services

During mobility, these applications may migrate between edge instances to preserve service continuity.

10.6 AI Computing Infrastructure

The AI server provides hardware acceleration for

- Deep Neural Networks
- Reinforcement Learning
- Time-Series Forecasting
- Mobility Prediction
- Traffic Classification
- Slice Optimization

GPU acceleration significantly reduces training and inference latency.

10.7 Monitoring Infrastructure

Continuous monitoring is essential for evaluating algorithm performance.

The monitoring subsystem records

- CPU Utilization
- Memory Usage
- Network Throughput
- Radio KPIs
- Packet Loss
- Latency
- Jitter
- SDR Status
- UE Statistics
- MEC Resource Usage

These measurements are visualized through Grafana dashboards.

10.8 Network Backbone

All laboratory equipment communicates through a centralized Layer-2/Layer-3 managed Ethernet switch.

The backbone provides

- 10 Gigabit Core Links
- Gigabit SDR Connections
- NAS Connectivity
- AI Workstation Access
- MEC Connectivity
- Monitoring Traffic

The backbone has been dimensioned to eliminate packet forwarding bottlenecks.

10.9 Synchronization Infrastructure

Accurate synchronization is one of the most important requirements of a real cellular network.

Although laboratory experiments can initially operate using internal oscillators, the proposed architecture supports future integration of

- GPS Disciplined Oscillator (GPSDO)
- IEEE 1588 Precision Time Protocol
- External 10 MHz Reference Clock
- Pulse Per Second (PPS)

These synchronization mechanisms improve frequency stability and enable precise timing during inter-cell mobility experiments.

10.10 Power Infrastructure

Continuous operation of the laboratory requires reliable electrical protection.

The power subsystem consists of

- Online UPS
- Rack Power Distribution Units
- Surge Protection

- Redundant Power Supplies
- Emergency Shutdown Controller

These systems protect sensitive RF and compute equipment against power interruptions.

10.11 Hardware Interaction

illustrates the interaction among all physical components.

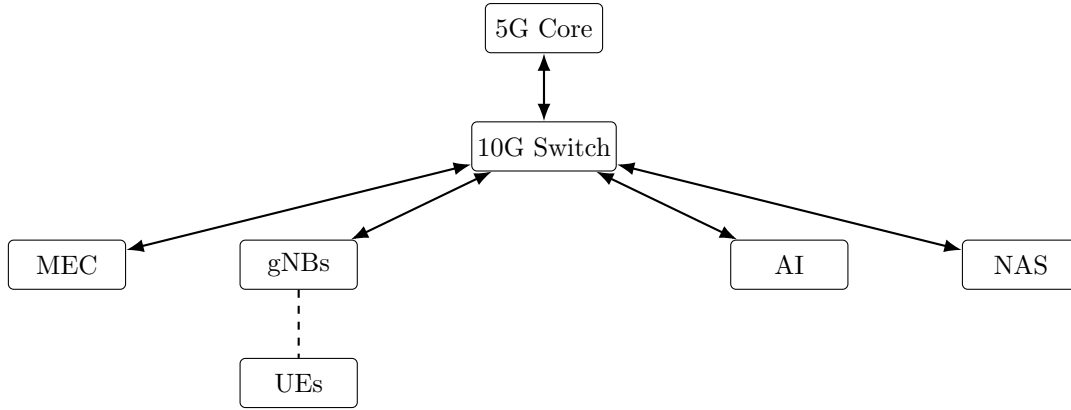


Figure 13: Interaction Between Major Hardware Components.

10.12 Scalability Considerations

The proposed hardware architecture is intentionally modular and supports incremental expansion.

Future upgrades may include

- Four or more gNodeBs
- Additional UPFs
- Distributed MEC clusters
- O-RAN Near-RT RIC
- Massive MIMO radios
- AI-native orchestration platforms
- 6G research extensions

The modular architecture ensures that future hardware additions can be incorporated without requiring significant redesign of the existing infrastructure.

11 Proposed Testbed Architecture

The proposed testbed is designed as a multi-tier private 5G network. It features standard 3GPP control interfaces alongside custom interfaces for AI control and telemetry collection.

11.1 Logical Architecture

illustrates the logical layout of the testbed, highlighting the interaction between the User Equipments, the SDR-based gNodeBs, the Open5GS Core Network, the MEC servers, and the AI Mobility Controller.

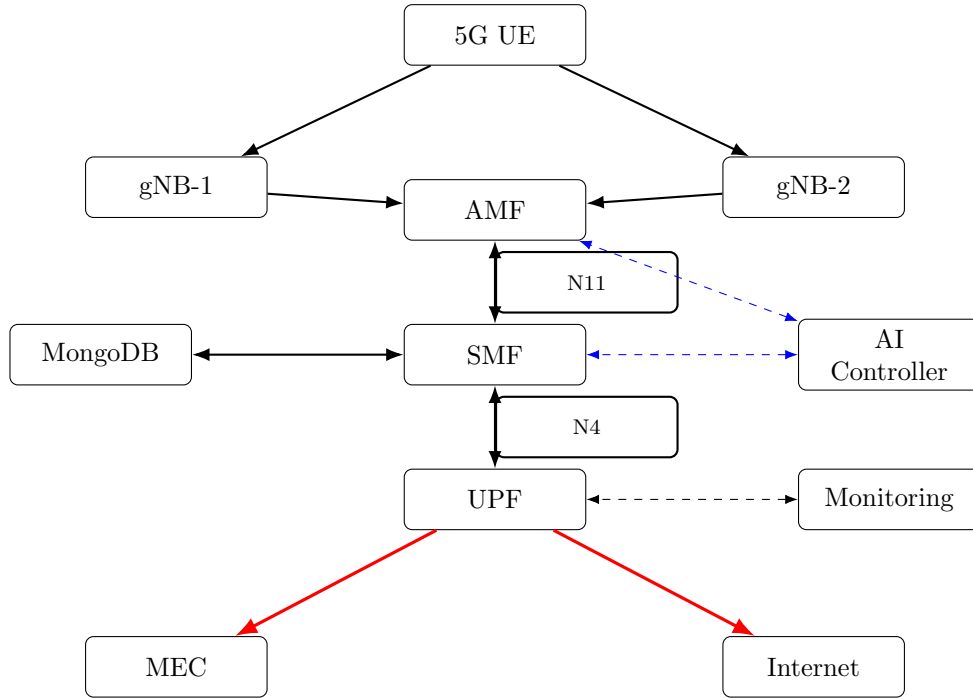


Figure 14: Logical Architecture of the Proposed AI-Assisted 5G Mobility Testbed.

11.2 End-to-End UE Registration Flow

11.3 Physical Deployment Topology

shows the physical deployment topology, illustrating how the hardware modules are interconnected using a Managed L2/L3 switch. All servers, workstations, and storage arrays are connected via high-speed Ethernet (1G/10G) to minimize control-plane latency.

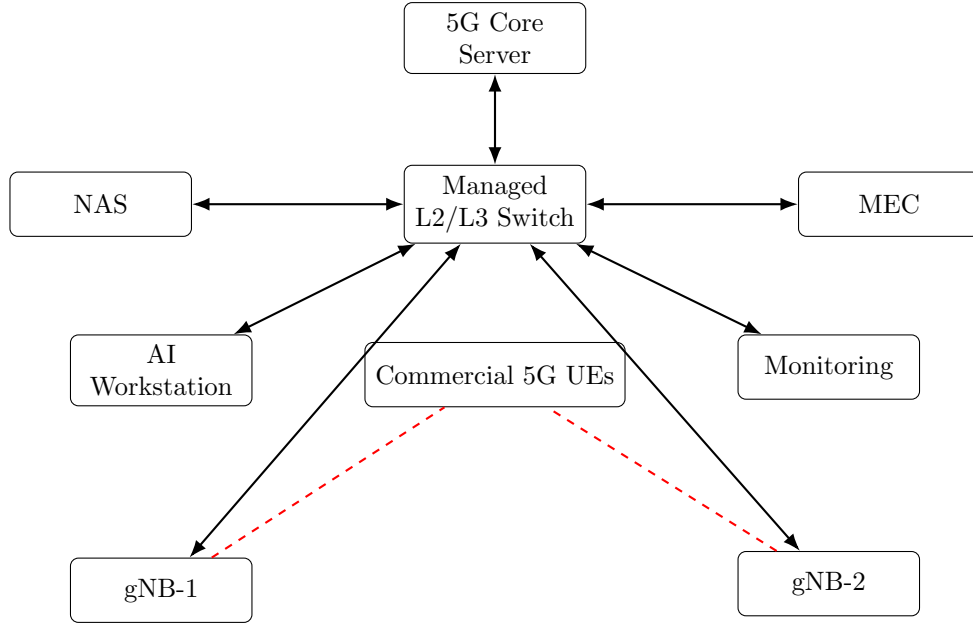


Figure 15: Physical Hardware Deployment Topology.

12 System Design and End-to-End Operational Architecture

The proposed testbed has been designed as a layered cyber-physical system that integrates radio access, core networking, edge computing, artificial intelligence, monitoring, and storage into a unified experimental platform. Unlike conventional laboratory deployments where individual subsystems operate independently, the proposed architecture enables continuous information exchange across every layer of the communication stack.

The complete system follows a closed-loop architecture in which network measurements continuously influence mobility decisions, resource allocation, network slice selection, and edge service placement.

12.1 Layered System Architecture

The proposed testbed is organized into seven logical layers.

12.2 Operational Workflow

The complete operation of the proposed system follows eight sequential stages.

1. User Equipment powers on.
2. UE registers with the nearest gNodeB.
3. Authentication is completed by the 5G Core.
4. A PDU Session is established.

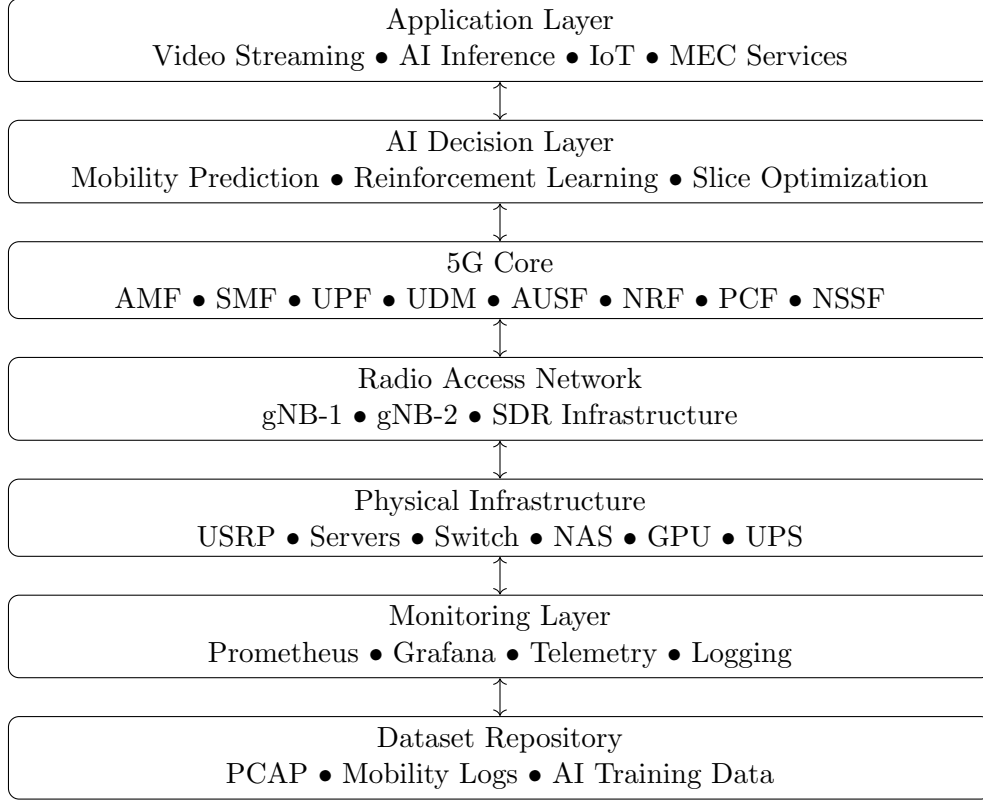


Figure 16: Layered Architecture of the Proposed Research Testbed

5. AI Mobility Controller begins collecting telemetry.
6. User starts moving across multiple cells.
7. AI predicts future mobility and prepares neighboring cells.
8. Handover completes with uninterrupted IP connectivity.

12.3 Closed-Loop System Operation

Unlike conventional cellular deployments that react to radio degradation, the proposed architecture continuously monitors the entire network.

The operational feedback loop consists of

12.4 End-to-End Data Flow

illustrates the complete path followed by signaling messages and user traffic throughout the proposed infrastructure.

12.5 Component Interaction Matrix

Table ?? summarizes the interaction between the major hardware and software components.

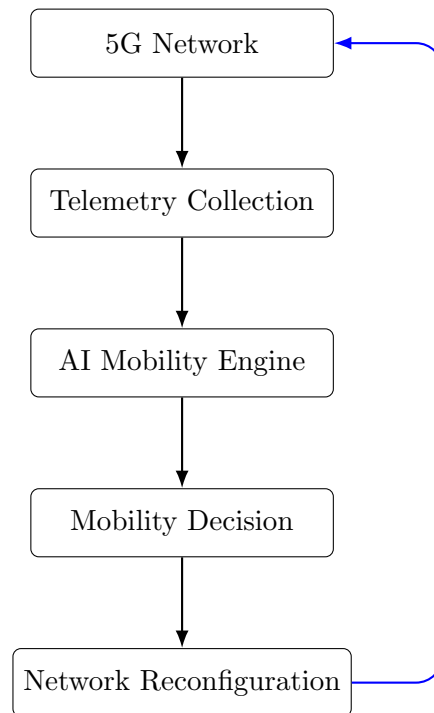


Figure 17: Closed-Loop AI-Assisted Mobility Optimization Framework.

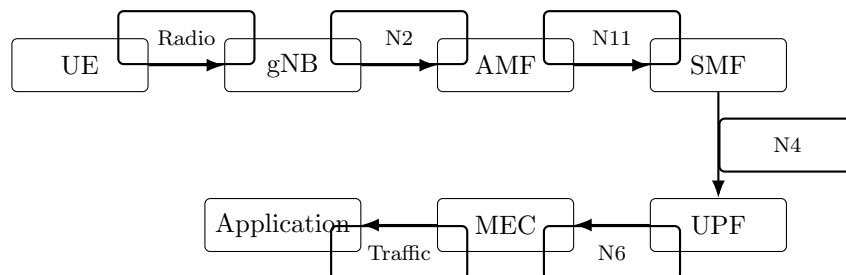


Figure 18: End-to-End Operational Data Flow.

12.6 Data Lifecycle

Every experiment conducted on the proposed platform generates multiple forms of telemetry.

The lifecycle of experimental data consists of

1. Packet Capture
2. KPI Collection
3. Time Synchronization
4. Storage
5. Dataset Construction
6. AI Training
7. Evaluation
8. Publication

This standardized pipeline ensures reproducibility of all experimental results and enables long-term dataset creation for future machine learning research.

12.7 System Design Advantages

Compared to existing software-only simulators, the proposed architecture provides

- Real radio propagation
- Physical SDR hardware
- Commercial User Equipments
- Standards-compliant 5G Core
- Edge Computing
- AI-assisted Mobility
- Network Slice Orchestration
- Continuous Telemetry
- Real-time Dataset Generation
- Future Open RAN compatibility
- Expandability toward AI-native 6G

Consequently, the proposed infrastructure serves as a complete experimental ecosystem capable of supporting both current 5G research and future Beyond-5G investigations.

13 Technical Justification of the Proposed Infrastructure

The successful realization of a realistic 5G Standalone research testbed requires considerably more than the deployment of a software-based core network and a pair of Software Defined Radios (SDRs). A complete experimental platform must reproduce the behaviour of commercial cellular infrastructure while providing sufficient flexibility to modify network functions, evaluate mobility algorithms, collect telemetry, and perform large-scale Artificial Intelligence (AI) experimentation.

Each hardware subsystem proposed in this project has therefore been selected based on its functional necessity rather than merely its computational capability. The following subsections justify the inclusion of every major infrastructure component within the proposed laboratory.

13.1 5G Core Server

The Core Server forms the central control entity of the private cellular network. It simultaneously hosts multiple Network Functions including the Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), Unified Data Management (UDM), Authentication Server Function (AUSF), Policy Control Function (PCF), Network Repository Function (NRF), and Network Slice Selection Function (NSSF).

A high-performance rack-mounted server is required because these services execute concurrently while processing large volumes of signaling traffic and user-plane packets. The use of enterprise-grade processors and high-speed storage minimizes processing latency during mobility procedures and ensures reliable long-duration operation.

13.2 Software Defined Radios

Unlike traditional networking experiments, mobility research requires actual radio transmission and reception. Software Defined Radios replace proprietary base station hardware and provide complete access to the physical layer.

The selected Ettus USRP B210 platform enables implementation of programmable gNodeBs supporting realistic radio transmission, scheduling, and mobility procedures. Two independent SDR platforms are required to establish overlapping radio cells and facilitate inter-cell handover experiments.

13.3 Commercial User Equipments

Software-based User Equipment simulators cannot accurately reproduce the radio characteristics, hardware timing, modem behaviour, and protocol implementations found in commercial devices.

Accordingly, the proposed laboratory employs commercial 5G smartphones operating in Standalone (SA) mode. These devices execute complete 3GPP procedures including registration, authentication, PDU session establishment, mobility management, and application-layer traffic generation.

13.4 Multi-access Edge Computing Server

Emerging 5G applications increasingly rely upon edge computing to reduce application latency. During user mobility, application services may require migration between edge servers while preserving session continuity.

A dedicated MEC server therefore enables investigation of edge-aware mobility management, service migration, traffic steering, and intelligent workload placement.

13.5 Artificial Intelligence Workstation

The proposed research incorporates Deep Learning and Reinforcement Learning models that continuously analyse network telemetry and predict future mobility events.

Training these models requires substantial computational resources including GPU acceleration, high-bandwidth memory, and fast storage. The AI workstation therefore performs both offline model training and online inference during experimental evaluation.

13.6 Monitoring Server

Objective evaluation of mobility algorithms requires continuous monitoring of every subsystem within the laboratory.

The monitoring server aggregates telemetry collected from the 5G Core, SDR hosts, User Equipments, AI controller, MEC server, and networking infrastructure. Prometheus exporters periodically collect system metrics, while Grafana dashboards provide real-time visualization of network behaviour throughout each experiment.

13.7 Network Attached Storage

Every mobility experiment generates significant volumes of packet captures, telemetry logs, application traces, and machine learning datasets.

A dedicated Network Attached Storage (NAS) appliance provides centralized, fault-tolerant storage for long-term archival and enables multiple researchers to access experimental datasets simultaneously without impacting operational servers.

13.8 Networking Infrastructure

The networking subsystem interconnects all compute nodes through a managed Layer-2/Layer-3 Ethernet switch supporting Gigabit and 10 Gigabit communication.

High-bandwidth switching minimizes packet forwarding latency between the Core Network, MEC server, AI workstation, and monitoring infrastructure. It additionally supports VLAN segmentation for isolating control-plane, user-plane, management, and storage traffic.

13.9 Power Infrastructure

Research experiments frequently execute continuously for several hours or even days. Unexpected power interruptions may invalidate experimental results and damage sensitive equipment.

The proposed Online UPS system provides uninterrupted electrical power, surge protection, and controlled shutdown capability during extended power failures, thereby preserving both experimental data and hardware reliability.

13.10 Infrastructure Integration

The proposed infrastructure has been designed according to three fundamental engineering principles:

1. Functional Isolation

Critical services such as the 5G Core, AI controller, MEC applications, monitoring platform, and storage infrastructure execute on dedicated hardware, eliminating resource contention.

2. Modular Expansion

Every subsystem communicates through standardized interfaces, allowing additional SDRs, MEC servers, User Equipments, and AI accelerators to be integrated without architectural redesign.

3. Research Flexibility

The use of programmable Software Defined Radios, open-source 5G software stacks, and containerized deployment enables rapid implementation of new mobility algorithms, network slicing techniques, and future Beyond-5G enhancements.

13.11 Overall Engineering Rationale

The selected infrastructure represents a balanced compromise between research capability, deployment complexity, and financial feasibility. Each subsystem directly supports one or more research objectives related to intelligent mobility management, AI-assisted handover optimization, network slicing, or edge computing.

Consequently, the proposed laboratory constitutes a reusable experimental platform capable of supporting multiple future research projects beyond the scope of the present proposal, including Open RAN, AI-native networking, Network Digital Twins, and sixth-generation (6G) communication systems.

14 Deployment Strategy and System Integration

Following procurement, the proposed testbed will be deployed in a phased manner to ensure incremental validation of every subsystem before full-scale experimental evaluation. Each deployment phase concludes with functional verification to isolate hardware, software, and networking issues before progressing to the next stage.

The deployment strategy minimizes integration risks while providing measurable milestones throughout laboratory establishment.

14.1 Phase I: Laboratory Preparation

Prior to hardware installation, the laboratory environment will be prepared to accommodate server infrastructure, networking equipment, and radio access hardware.

Activities include

- Installation of equipment rack
- Electrical wiring
- Dedicated UPS connection
- Air conditioning verification
- Ethernet backbone installation
- Cable management
- Internet uplink provisioning

14.2 Phase II: Compute Infrastructure Deployment

The compute layer forms the foundation of the laboratory.

The following systems are installed and verified.

- Core Network Server
- MEC Server
- AI Workstation
- Monitoring Server
- NAS Storage

Each system undergoes

- BIOS validation
- RAID configuration

- Operating System installation
- Network configuration
- SSH connectivity verification
- Time synchronization

14.3 Phase III: Network Infrastructure Deployment

The networking layer interconnects all compute nodes.

Deployment activities include

- Managed switch configuration
- VLAN creation
- IP address assignment
- 10 Gigabit uplink configuration
- Routing verification
- Firewall configuration

14.4 Phase IV: 5G Core Deployment

The software components of the 5G Core are deployed using containerized services.

The deployment sequence consists of

1. MongoDB
2. NRF
3. UDM
4. AUSF
5. AMF
6. SMF
7. PCF
8. NSSF
9. UPF

Following deployment, all Service Based Interfaces (SBIs) are verified using health checks and functional tests.

14.5 Phase V: Radio Access Network Deployment

Each Software Defined Radio is connected to its host workstation.

Deployment includes

- UHD Driver installation
- SDR firmware update
- Clock synchronization
- RF calibration
- gNodeB configuration
- Cell broadcast verification

Commercial User Equipments are then attached to the private network for initial registration testing.

14.6 Phase VI: Monitoring and Telemetry Deployment

Monitoring infrastructure is activated prior to experimental execution.

Telemetry sources include

- Core Network metrics
- SDR statistics
- Operating system statistics
- GPU utilization
- MEC resource usage
- Network bandwidth
- Packet captures

Grafana dashboards are configured to provide real-time visibility into every subsystem.

14.7 Phase VII: AI Framework Deployment

After successful validation of the baseline network, the Artificial Intelligence framework is introduced.

Deployment activities include

- Dataset collection

- Feature extraction
- Model training
- Model validation
- Online inference deployment
- Closed-loop testing

14.8 System Integration Workflow

illustrates the complete deployment pipeline.

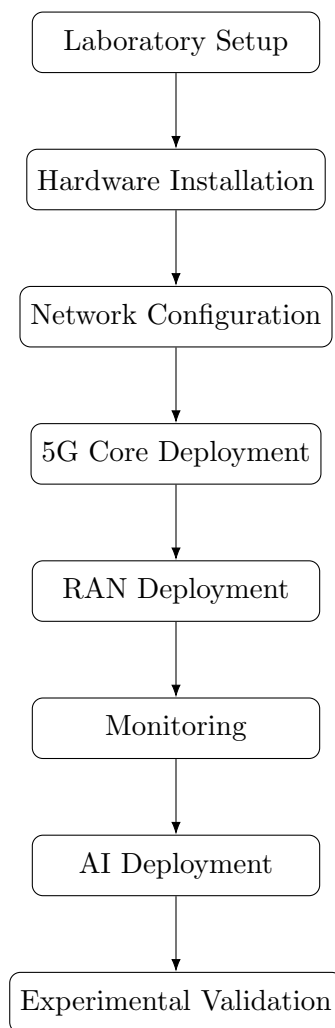


Figure 19: Phased Deployment and Integration Workflow

14.9 Verification Checklist

Each deployment phase concludes with predefined acceptance tests.

14.10 Expected Deployment Duration

The complete infrastructure is expected to be operational within approximately twelve weeks following procurement.

15 Required Modules and Procurement Matrix

The hardware requirements for the proposed 5G testbed are structured into nine distinct modules. Table provides a detailed procurement matrix outlining the hardware components, specifications, target pricing, and verified vendors in India.

Table 12: Procurement and Module Matrix for the Proposed 5G Standalone Testbed

Subsystem	Module	Key Technical Specifications	Qty	Procurement Source	Estimated Cost (INR)
1. 5G Core	Dell PowerEdge R760 Rack Server	Dual Intel Xeon Silver 4410Y, 128 GB DDR5 RAM, 2 × 480 GB SSD (RAID-1), Dual 10 GbE SFP+	1	Dell India / ServerBasket India	3,50,000
2. 5G RAN	Ettus USRP B210 Software Defined Radio	70 MHz–6 GHz, 2×2 MIMO, USB 3.0, SMA RF Interface, External Omni-directional Antennas	2	GSAS Micro Systems Bangalore / Element14 India	5,00,000
3. User Equipment	Commercial Android 5G Smartphones	5G Standalone (SA) Support, NR Band n78, Manual PLMN Selection	5	Amazon India / Samsung / OnePlus	1,50,000
4. MEC Server	HPE ProLiant DL360 Gen11	Intel Xeon Silver, 64 GB DDR5 RAM, 1.92 TB NVMe SSD, Dual 10 GbE NIC	1	HPE India Partner / Cyberwala	2,50,000

Continued on Next Page

Table 12 Continued from Previous Page

Subsystem	Module Details	Key Specifications	Technical	Qty	Procurement Source	Estimated Cost (INR)
5. AI Workstation	Custom Liquid-Cooled GPU Workstation	Intel Core i9-14900K, NVIDIA RTX 4090 (24 GB), 64 GB DDR5, 2 TB Gen4 NVMe SSD		1	Volted PC Delhi / PC Kumar Bangalore	3,50,000
6. Monitoring Server	Dell PowerEdge R250	Intel Xeon E-2314, 32 GB RAM, Dual 1 TB HDD, Dual Gigabit Ethernet		1	Dell India / ServerBasket India	80,000
7. NAS Storage	Synology DS1821+ NAS	8-Bay NAS with 4 × 4 TB Seagate Iron-Wolf HDDs configured in RAID-5		1	Computech Store / Vishal Peripherals	1,40,000
8. Networking	Managed L2/L3 Enterprise Switch	24-Port Gigabit PoE+, 4 × 10 Gb SFP+ Uplinks, Cat6A Structured Cabling		1	Vishal Peripherals / L&T SuFin	60,000
9. Power Infrastructure	APC Smart-UPS RC 5 kVA	Online Double Conversion UPS with External Battery Cabinet (Approx. 30 Minutes Backup)		1	APC Schneider Electric India / Vertiv India	1,20,000
Total Estimated Capital Expenditure (CAPEX)						20,00,000

16 Detailed Module Technical Specifications & Visualizations

This section provides technical descriptions and visual references for the key physical hardware modules to be procured for the laboratory.

16.1 Module 1: Software-Defined Radio (SDR) RAN Transceiver

The Radio Access Network (RAN) is implemented using the Ettus Research USRP B210. This platform is chosen for its native integration with open-source LTE/5G stacks (srsRAN and OpenAirInterface). It operates as a fully flexible gNodeB base station.

- **Frequency Range:** 70 MHz to 6 GHz (supporting 5G SA FR1 bands, specifically n78).
- **RF Channels:** 2×2 MIMO transceiver offering up to 56 MHz of real-time bandwidth.
- **Interface:** High-speed USB 3.0 connection to the host PC running srsRAN gNB software.

shows the physical Ettus USRP B210 unit in its metal enclosure, with RF cabling and SMA antennas connected on a laboratory workbench.



Figure 20: Ettus Research USRP B210 SDR Unit (Procured via GSAS Micro Systems)

16.2 Module 2: 5G Core Network Rack Server

The Core Network runs Open5GS. It requires a dedicated, reliable rack-mounted server capable of handling hundreds of subscriber sessions, managing control-plane database queries, and routing high-throughput user-plane traffic (UPF) with low latency.

- **Model:** Dell PowerEdge R760 (2U Rack Server).
- **Compute:** Dual Intel Xeon Silver processors with hardware virtualization enabled.
- **Connectivity:** Intel X550 Dual Port 10GbE SFP+ Network Interface Card for low-latency routing to the MEC server and managed switches.

illustrates the front interface of the Dell PowerEdge R760 server showcasing the hot-swappable drive bays and indicator LEDs.



Figure 21: Dell PowerEdge R760 5G Core Server (Procured via Dell India / ServerBasket)

16.3 Module 3: AI and Mobility Controller Workstation

The AI layer hosts the DNN-based slice selection algorithms, traffic prediction models, and deep reinforcement learning (DRL) agents. Training and running these models in real time during handover events requires high VRAM and parallel compute capabilities.

- **Compute:** Intel Core i9-14900K with 64GB DDR5 RAM.
- **Accelerator:** NVIDIA GeForce RTX 4090 with 24GB GDDR6X VRAM, optimized for CUDA and TensorRT inference.
- **Cooling:** Custom closed-loop liquid cooling system to prevent thermal throttling during continuous training cycles.

shows the physical design of the liquid-cooled AI workstation, featuring multiple GPU configurations and custom cooling tubes.



Figure 22: High-Performance Liquid-Cooled AI GPU Workstation (Procured via Volted PC / PC Kumar)

16.4 Module 4: Central Storage NAS Chassis

Network logs, Wireshark packet captures (PCAPs), and telemetry metrics from Grafana/Prometheus must be centralized and backed up for training dataset creation. This is handled by a high-availability network storage unit.

- **Model:** Synology DS1821+ (8-Bay Desktop NAS).

- **Storage:** $4 \times 4\text{TB}$ Seagate IronWolf NAS-grade SATA HDDs configured in RAID 5.
- **Expansion:** Built-in PCIe slot populated with a dual-port 10GbE card for high-speed file transfers.

displays the front and rear panels of the Synology DS1821+ NAS unit.



Figure 23: Synology DS1821+ 8-Bay NAS Chassis (Procured via Computech / Vishal Peripherals)

17 Software Stack Integration Plan

The testbed leverages standard open-source cellular stacks integrated with standard container orchestration. outlines the software layers implemented across the physical servers.

- **5G Core Network (Open5GS):** Provides AMF (Access and Mobility Management), SMF (Session Management), UPF (User Plane Function), UDM (Unified Data Management), and AUSF (Authentication Server Function). It uses MongoDB to store subscriber profiles (SIM details, slice subscription, keys).
- **Radio Access Network (srsRAN Project):** Instantiates a software-based gNodeB that interfaces with the USRP B210 via the UHD (USRP Hardware Driver) API. It supports split-8 PHY/MAC layers.
- **Edge Computing Platform (Kubernetes/Docker):** Deployed on the MEC server to manage containerized edge services. KubeEdge or custom load-balancers redirect traffic dynamically during handovers.
- **Telemetry Stack (Prometheus & Grafana):** Continuously queries cellular KPIs (e.g., active UEs, bandwidth per slice, packet latency, jitter, RSRP, and drop rates) using custom exporters on Open5GS and srsRAN.

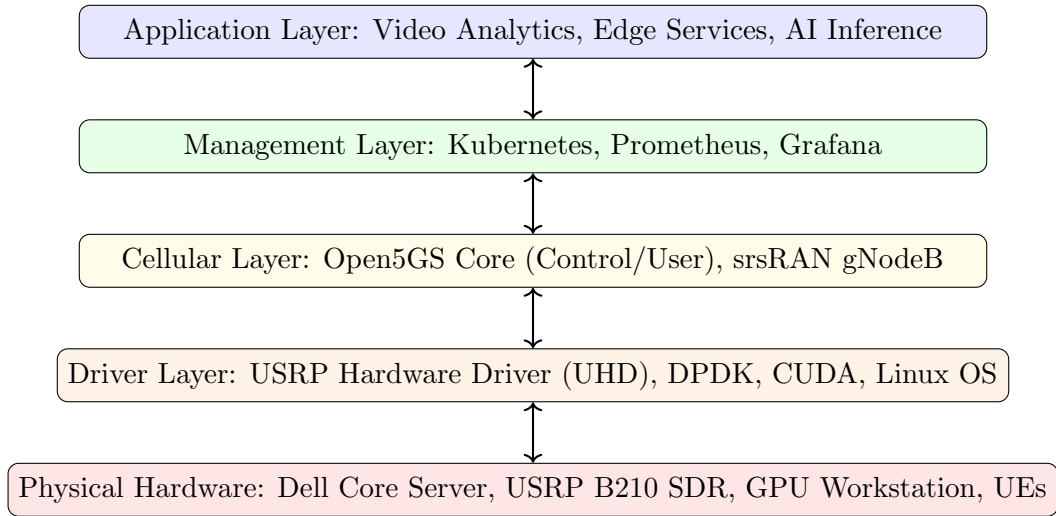


Figure 24: 5G Testbed Integrated Software Stack

18 Junior Research Fellow (JRF) Profile

A Junior Research Fellow (JRF) will be recruited to support the installation, integration, and day-to-day operation of the 5G testbed.

18.1 Essential Technical Qualifications

- B.Tech/M.Tech in Computer Science, Electronics & Communication, or Information Technology.
- **Linux Systems Administration:** Experience with Ubuntu Server, systemd, bash scripting, network interface configurations (IP routing, VLANs, bridging).
- **Programming:** Proficiency in Python and C/C++.
- **Networking Fundamentals:** Deep understanding of the TCP/IP stack, routing protocols, sockets, and network analysis tools (Wireshark, tcpdump).

18.2 Desirable Technical Qualifications

- GATE qualification in CS or EC.
- Hands-on experience compiling and configuring Open5GS, srsRAN, or OpenAirInterface.
- Familiarity with Software-Defined Radio (SDR) hardware and UHD library.
- Knowledge of Docker, Kubernetes, and containerized deployment patterns.
- Basic understanding of machine learning frameworks (PyTorch or TensorFlow).

19 Risk Assessment & Mitigation Plan

Table ?? lists potential technical and logistical risks associated with the implementation of the 5G testbed, along with corresponding mitigation strategies.

Table 2: Major Interfaces in the Proposed 5G Standalone Testbed

Interface	Connected ties	Enti-	Purpose
midrule N1	UE $\leftrightarrow$ AMF		NAS signalling, registration, authentication, mobility management, security context establishment.
N2	gNB $\leftrightarrow$ AMF		Control-plane signalling between the Radio Access Network and the Core Network including registration requests, paging, handover commands, and UE context transfer.
N3	gNB $\leftrightarrow$ UPF		User-plane interface responsible for transporting GTP-U encapsulated user traffic between the Radio Access Network and the User Plane Function.
N4	SMF $\leftrightarrow$ UPF		PCFP signalling used to configure forwarding rules, establish GTP tunnels, and update user-plane routes during mobility events.
N6	UPF $\leftrightarrow$ Data Network		Interface connecting the 5G Core to external IP networks, cloud services, enterprise networks, and the Internet.
N8	UDM $\leftrightarrow$ AUSF		Authentication credential exchange and subscriber identity verification.
N10	SMF $\leftrightarrow$ UDM		Retrieval of subscriber session information and service subscriptions.
N11	AMF $\leftrightarrow$ SMF		Session establishment requests, PDU session management, and mobility-related signalling.
N22	AMF $\leftrightarrow$ NSSF		Selection of appropriate Network Slice Instances based on subscriber profile and requested services.
Xn	gNB $\leftrightarrow$ gNB		Direct communication between neighbouring base stations for fast inter-cell handover and context transfer.
SBI	Core Network Functions		RESTful HTTP/2 Service-Based Interfaces enabling communication among AMF, SMF, NRF, UDM, PCF, AUSF, and NSSF.

Table 3: Input Features Used by the AI Mobility Controller

Category	Collected Parameters
Radio Measurements	RSRP, RSRQ, SINR, CQI, BLER
Mobility	Velocity, Direction, GPS Coordinates, Cell Residence Time
Core Network	AMF Load, SMF Load, Active Sessions
User Plane	Throughput, Packet Delay, Packet Loss, Jitter
Edge Computing	CPU Utilization, GPU Utilization, Memory Usage
Network Slicing	Slice Availability, Slice Utilization, Reserved Resources
Historical Data	Past Mobility Patterns, Previous Handover Success Rate

Table 4: Baseline Algorithms

Algorithm	Description
RSRP Based Handover	Conventional strongest-signal selection.
RSRP + Hysteresis	Standard LTE/5G implementation using configurable hysteresis margins.
A3 Event Trigger	3GPP Event A3 mobility procedure.
Time-to-Trigger (TTT)	Threshold-based delayed handover.
Proposed AI Mobility	Deep Reinforcement Learning assisted predictive mobility management.

Table 5: Mobility Test Scenarios

Scenario	Description	Velocity
Pedestrian	Walking user	3–5 km/h
Cyclist	Campus mobility	15 km/h
Vehicle	Urban road	30–60 km/h
High Speed	Fast moving UE	80–120 km/h
Dense Cell	Frequent handovers	Variable

Table 6: Evaluation Metrics

Metric	Purpose
Handover Delay	Time required to complete mobility.
Packet Loss	Packets dropped during mobility.
Latency	End-to-end network delay.
Jitter	Variation in packet delay.
Throughput	Application data rate.
RSRP	Signal strength.
SINR	Signal quality.
CQI	Channel quality.
Slice Utilization	Resource utilization.
MEC Response Time	Application latency.
CPU Usage	Core server utilization.
GPU Usage	AI inference workload.
Power Consumption	Energy efficiency.

Table 7: Planned Experimental Campaign

Phase	Objective
Phase I	Deployment and validation of the private 5G Standalone network.
Phase II	Baseline mobility experiments using standard 3GPP algorithms.
Phase III	Collection of mobility telemetry datasets.
Phase IV	Training of Deep Learning and Reinforcement Learning models.
Phase V	Deployment of AI-assisted mobility controller.
Phase VI	Comparative evaluation under multiple mobility scenarios.
Phase VII	Scalability analysis involving increasing numbers of users and cells.
Phase VIII	Publication, dataset release, and technology demonstration.

Table 8: Hardware Classification of the Proposed Testbed

Category	Purpose
5G Core Infrastructure	Implements all Core Network Functions.
Radio Access Network	Implements gNodeBs using SDRs.
Commercial User Equipments	Generate real mobility traffic.
Edge Computing Infrastructure	Runs MEC applications.
AI Computing Platform	Hosts prediction and reinforcement learning models.
Storage Infrastructure	Stores telemetry and datasets.
Monitoring Infrastructure	Collects system metrics.
Networking Infrastructure	Provides low-latency Ethernet connectivity.
Synchronization Infrastructure	Provides accurate timing and clock synchronization.
Power Infrastructure	Ensures uninterrupted operation.

Table 9: Major Component Interactions

Source	Destination	Purpose
UE	gNB	Radio Access
gNB	AMF	Registration
AMF	UDM	Authentication
AMF	SMF	Session Management
SMF	UPF	Tunnel Creation
UPF	MEC	Edge Traffic
UPF	Internet	External Access
Prometheus	AI	Telemetry
AI	AMF	Mobility Decision
AI	SMF	Slice Decision
AI	MEC	Service Migration
Grafana	Operators	Monitoring

Table 10: Deployment Verification Checklist

Subsystem	Verification
Core Network	Successful UE Registration
RAN	Cell Broadcast Detection
UPF	Internet Connectivity
MEC	Application Deployment
AI Controller	Real-Time Inference
Monitoring	Live Dashboard
NAS	Log Archival
Mobility	Successful Handover

Table 11: Estimated Deployment Timeline

Activity	Duration
Laboratory Preparation	2 Weeks
Hardware Installation	2 Weeks
Network Configuration	1 Week
5G Core Installation	2 Weeks
RAN Deployment	2 Weeks
AI Integration	2 Weeks
System Validation	1 Week

Table 13: Project Risk Assessment and Mitigation Matrix

Risk Event		Impact Level	Mitigation Strategy
Procurement Delays	De-	Medium	Utilize pre-approved Indian distributors (GSAS India, Dell India, HPE India) to minimize customs clearance delays and ensure timely delivery.
SDR Overheating	Hardware	Medium	Install active cooling fans on the USRP B210 enclosures, provide adequate ventilation, and limit continuous high-power transmission during prolonged experiments.
RF Interference		High	Conduct initial experiments using RF coaxial cables with programmable attenuators before transitioning to over-the-air testing. Employ spectrum analysis to identify and mitigate interference sources.
Core Configuration Errors	Network	High	Maintain version-controlled configuration files, periodic VM snapshots, Docker container backups, and automated health checks for rapid recovery.
UPS Battery Failure	Fail-	Low	Establish preventive maintenance contracts with the UPS vendor, perform periodic battery health monitoring, and configure graceful server shutdown using network-enabled UPS management.